Debugging

You can connect the debugger in your editor, for example with Visual Studio Code or PyCharm.

Call `uvicorn`

In your FastAPI application, import and run `uvicorn` directly:

Python 3.10+

```python
import uvicorn
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
def root():
    a = "a"
    b = "b" + a
    return {"hello world": b}


if __name__ == "__main__":
    uvicorn.run(app, host="0.0.0.0", port=8000)
```

About `__name__ == "__main__"`

The main purpose of the `__name__ == "__main__"` is to have some code that is executed when your file is called with:

```bash
fast →
$ python myap
```

but is not called when another file imports it, like in:

```python
from myapp import app
```

More details

Let's say your file is named `myapp.py`.

If you run it with:

Ask AI

```bash
fast →
$ python myap
```

then the internal variable `__name__` in your file, created automatically by Python, will have as value the string `"__main__"`.

So, the section:

```
uvicorn.run(app, host="0.0.0.0", port=8000)
```

will run.

This won't happen if you import that module (file).

So, if you have another file `importer.py` with:

```
from myapp import app

# Some more code
```

in that case, the automatically created variable `__name__` inside of `myapp.py` will not have the value `"__main__"`.

So, the line:

```
uvicorn.run(app, host="0.0.0.0", port=8000)
```

will not be executed.

> **Info**
>
> For more information, check the official Python docs.

Run your code with your debugger

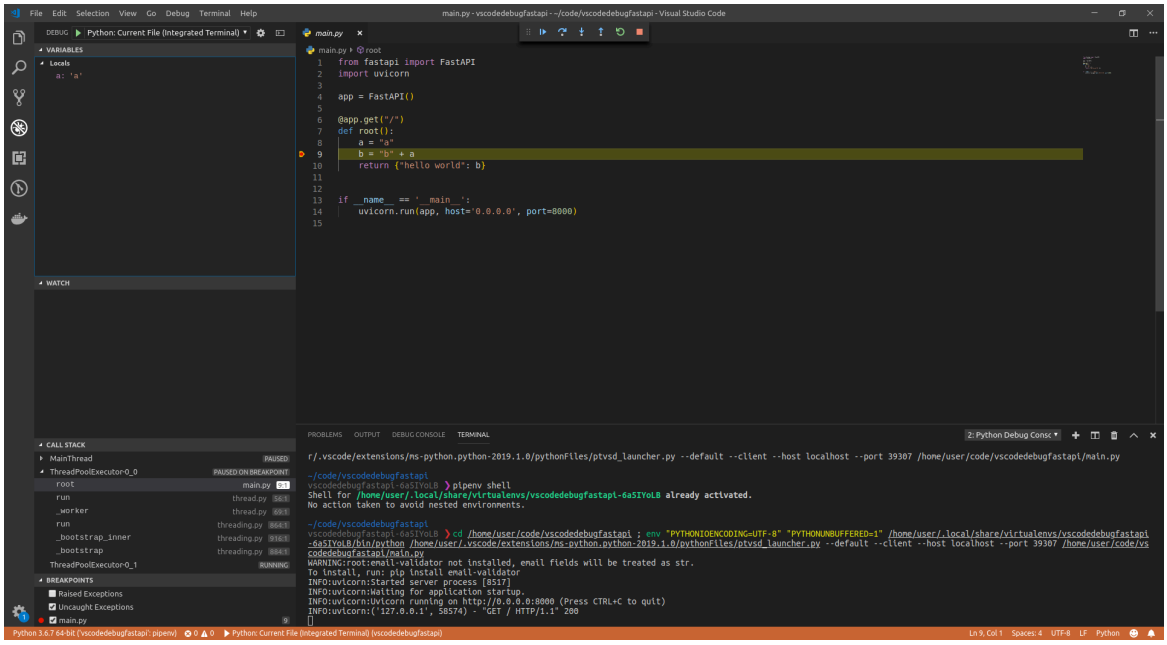
Because you are running the Uvicorn server directly from your code, you can call your Python program (your FastAPI application) directly from the debugger.

For example, in Visual Studio Code, you can:

- Go to the "Debug" panel.
- "Add configuration...".
- Select "Python"
- Run the debugger with the option "`Python: Current File (Integrated Terminal)`".

It will then start the server with your **FastAPI** code, stop at your breakpoints, etc.

Here's how it might look:

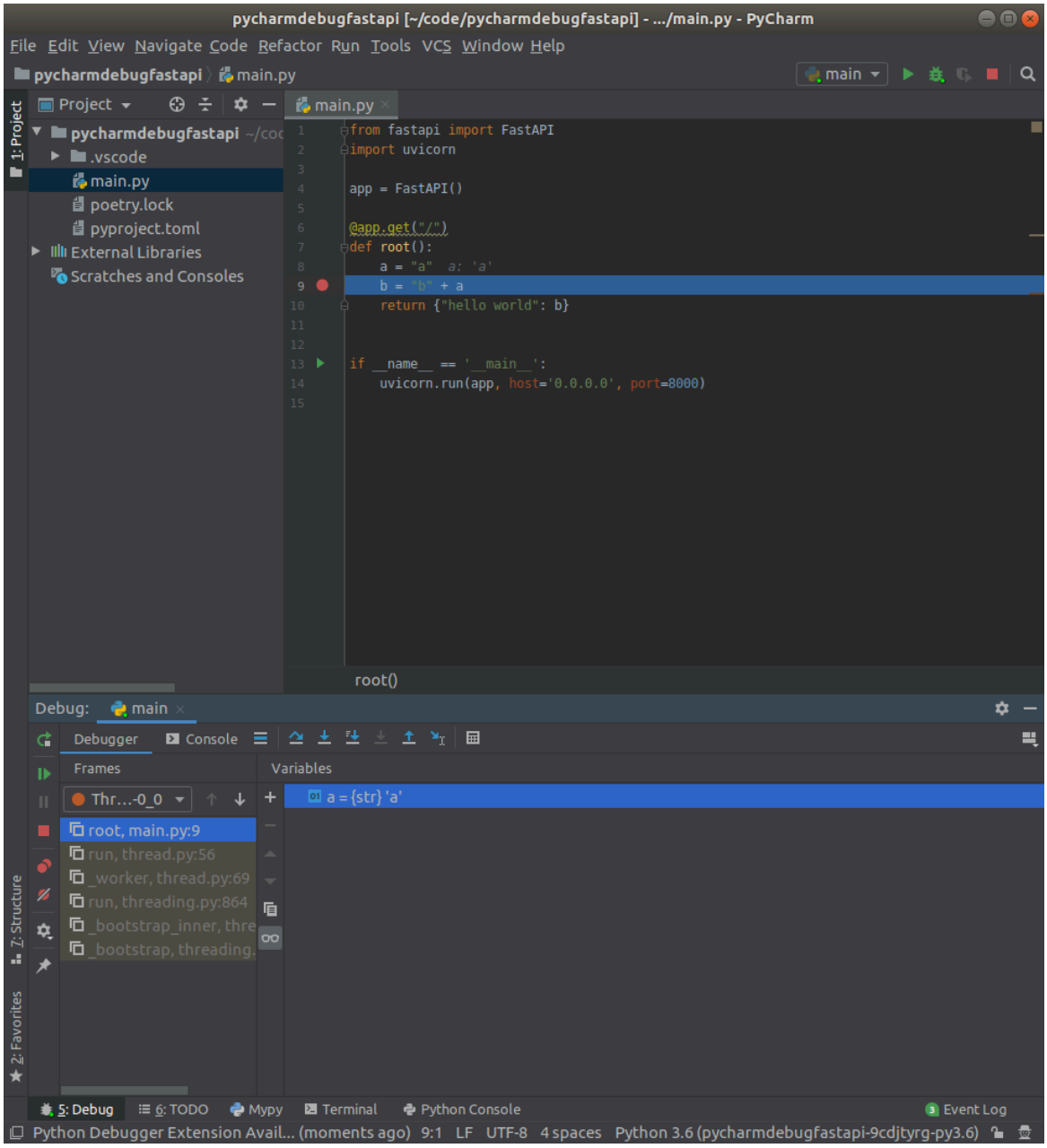


If you use Pycharm, you can:

- Open the "Run" menu.
- Select the option "Debug...".
- Then a context menu shows up.
- Select the file to debug (in this case, `main.py`).

It will then start the server with your **FastAPI** code, stop at your breakpoints, etc.

Here's how it might look:


pycharmdebugfastapi [~/code/pycharmdebugfastapi] - .../main.py - PyCharm
File Edit View Navigate Code Refactor Run Tools VCS Window Help
pycharmdebugfastapi main.py
main
Project
pycharmdebugfastapi ~/coc
.vscode
main.py
poetry.lock
pyproject.toml
External Libraries
Scratches and Consoles
main.py
from fastapi import FastAPI
import uvicorn
app = FastAPI()
@app.get("/")
def root():
a = "a" a: 'a'
b = "b" + a
return {"hello world": b}
if __name__ == '__main__':
uvicorn.run(app, host='0.0.0.0', port=8000)
root()
Debug: main
Debugger Console
Frames
Variables
Thr...-0_0
root, main.py:9
run, thread.py:56
_worker, thread.py:69
run, threading.py:864
_bootstrap_inner, thre
_bootstrap, threading.
a = {str} 'a'
5: Debug 6: TODO Mypy Terminal Python Console Event Log
Python Debugger Extension Avail... (moments ago) 9:1 LF UTF-8 4 spaces Python 3.6 (pycharmdebugfastapi-9cdjtyrg-py3.6)